World Patent Information 85 (2026) 102450

Contents lists available at ScienceDirect

World Patent Information

journal homepage: www.elsevier.com/locate/worpatin

Review

Beyond citations: A dynamic semantic graph approach for early-stage patent valuation

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ARTICLE INFO

Keywords:
Early-stage patent valuation
Graph neural networks (GCN)
Natural Language Processing (NLP)
Dynamic thresholding

ABSTRACT

Background & objective: In the knowledge-driven economy, identifying high-potential inventions immediately upon disclosure is critical for strategic R&D management. Traditional valuation methods predominantly rely on forward citations, which suffer from "time-lag biases," rendering them ineffective for assessing newly published patents. Furthermore, conventional AI-based approaches often treat patents as isolated textual entities, overlooking their latent topological relationships. This study proposes a "zero-lag" framework for early-stage patent valuation by introducing **Strategic Topological Value (STV)**—a metric that quantifies an invention's relational significance and structural importance within a technological network.
Methodology: We develop an Integrated Multimodal Deep Learning Framework that fuses Sentence-BERT (SBERT) embeddings with Graph Convolutional Networks (GCN). To overcome the data sparsity of citation networks, we implement a Dynamic Thresholding Graph Construction algorithm to reconstruct a "Latent Semantic Graph." Based on a global random sampling from the entire population of 233,122 patents in CPC Section H (Electricity), and guided by a sensitivity analysis across $K \in [5,30]$, a K-nearest neighbor ($K = 25$) approach is utilized to convert unstructured text into a structured network where edges represent intrinsic technological proximity. Patents are classified into High-STV and Low-STV categories based on their structural centrality within this latent topology.
Empirical results: Rigorous evaluation was conducted through 30 independent experimental runs on samples from Section H domain. The results demonstrate that the proposed GCN model achieves a robust Mean Accuracy of 0.955 ($\pm$0.006), an F1-score of 0.956 ($\pm$0.006), and an AUC of 0.993 ($\pm$0.002). This performance significantly outpaces traditional text-only baselines, including Multi-Layer Perceptron (Accuracy: 0.825 $\pm$ 0.011) and Random Forest (Accuracy: 0.794 $\pm$ 0.013), providing strong empirical validation for the Structural Positioning advantage in high-tech patent valuation.
Conclusion & practical implications: By shifting the valuation paradigm from "content-only" analysis to "structural positioning," this research provides a proactive decision-support tool. The high computational efficiency (averaging 21.68 s per batch on an RTX 4060 GPU) confirms its feasibility for real-time, large-scale patent analytics, enabling organizations to identify "hidden gems" immediately upon publication without the multi-year waiting period required for citation maturation.

1. Introduction

In the contemporary knowledge-driven economy, Intellectual Property (IP), particularly patents, has evolved into a strategic core asset that dictates corporate competitive advantage and national innovation capacity [1,2]. As global R&D expenditures continue to surge, the volume of patent applications has reached unprecedented levels—a phenomenon often referred to as "Big Patent Data." For R&D managers and Chief Technology Officers (CTOs), the critical challenge has shifted from accessing data to filtering noise: How to efficiently identify high-value patents—often termed "hidden gems"—from millions of low-quality filings? Effectively valuing these assets is no longer just a legal necessity but a fundamental component of strategic R&D management and technology futures analysis [3].

Despite this urgency, patent valuation remains a notoriously difficult task. Traditional approaches predominantly rely on bibliometric

This article is part of a special issue entitled: AI/ML IP data analytics for innovation published in World Patent Information.

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https://doi.org/10.1016/j.wpi.2026.102450
Received 31 December 2025; Received in revised form 26 February 2026; Accepted 2 March 2026
Available online 10 March 2026

indicators, such as forward citations and patent family size [4]. While statistically robust for historical analysis, these metrics suffer from a severe "Time-Lag Bias" [5]. A breakthrough innovation typically requires three to five years to accumulate sufficient citations to be recognized as "valuable" [6]. This delay renders citation-based metrics ineffective for assessing newly granted or pending patents. To bridge this gap, recent advancements in Artificial Intelligence (AI) have introduced Natural Language Processing (NLP) techniques to extract semantic features from patent text [7,8]. Furthermore, as Rettenmeier et al. [9] demonstrated, systematic patent analysis is instrumental in identifying specific technological trends and forecasting innovation trajectories.

However, a fundamental methodological limitation persists: most existing AI-based approaches treat patents as Independent and Identically Distributed (IID) text documents. This approach is essentially "topology-blind," as it overlooks the core principle of Recombinant Innovation—the theory that a patent's value is determined not merely by its internal technical disclosure (what it says) but by its Structural Positioning within the broader landscape of technological knowledge (where it stands). By analyzing patents in isolation, traditional NLP models—including Multi-Layer Perceptron (MLP) and Random Forest (RF) baselines—fail to capture the relational advantages and connectivity embedded in the technological network.

To address these limitations, this study proposes an Integrated Multimodal Deep Learning Framework that shifts the focus from "content-only" to "content-in-context." We introduce the concept of Strategic Topological Value (STV), a metric that quantifies an invention's relational significance within a dynamic knowledge network. To operationalize this, we develop a Dynamic Thresholding Graph Construction mechanism that reconstructs a "Latent Semantic Graph" by calculating high-dimensional similarity between patent abstracts using Sentence-BERT (SBERT). Unlike previous studies that rely on sparse citation networks, our algorithm effectively converts unstructured text into a structured network where edges represent intrinsic technological proximity. Based on an empirical sensitivity analysis conducted across $K \in [5, 30]$, we utilize a K-nearest neighbor ($K = 25$) approach to ensure the robustness of the semantic topology. We then employ Graph Convolutional Networks (GCN) to model the valuation problem. Through a neighborhood aggregation (message-passing) mechanism, the GCN explicitly learns the topological importance of a patent, allowing for the quantification of its STV as a leading indicator of value immediately upon publication.

This research offers three primary contributions to the field of IP Data Analytics. First, we demonstrate a Dynamic Graph Construction strategy that addresses the sparsity and over-smoothing problems common in graph learning, ensuring that GCNs can be robustly applied to large-scale patent datasets without relying on explicit citation links. Second, we provide rigorous empirical evidence of the decisive role of structural topology. Through 30 independent experimental runs utilizing global random sampling from the BigPatent dataset [10], specifically targeting the entire population of 233,122 CPC Section H (Electricity) patents, our results show that the proposed GCN model achieves a robust Mean Accuracy of 0.955 ($\pm$0.006) and an AUC of 0.993 ($\pm$0.002). This significantly outperforms text-only baselines such as MLP (0.825 $\pm$ 0.011) and RF (0.794 $\pm$ 0.013), confirming that Strategic Topological Value (STV) is a more powerful predictor of patent impact than textual content alone. Finally, we provide a "Zero-Lag" valuation tool for managerial utility. By utilizing semantic topology, R&D managers and CTOs can assess the potential impact of patents immediately upon publication, facilitating proactive portfolio optimization, M&A due diligence, and competitor intelligence.

2. Literature review and related work

This section reviews the theoretical evolution of patent valuation, transitioning from traditional econometric indicators to modern AI-driven approaches. We categorize existing literature into three distinct streams: (1) Bibliometric and citation-based valuation methods; (2) Semantic analysis using Natural Language Processing (NLP); and (3) The emerging paradigm of network-based representation learning and Graph Neural Networks (GNN).

2.1. The evolution of patent value indicators: from static to predictive analytics

The economic valuation of patents has historically relied on the premise that patent statistics reflect innovative output. Hall et al. [4] established the foundational framework linking patent metrics—such as forward citations, patent family size, and renewal data—to firm market value. Specifically, forward citations have long been regarded as the "gold standard" for assessing technological impact, a correlation confirmed by Harhoff et al. [6]. Existing literature has extensively refined these metrics; for instance, Aristodemou and Tietze [1] provided a comprehensive taxonomy of Intellectual Property (IP) analytics, while Reitzig [2] identified key determinants of patent value at the granting stage.

Recent research in *World Patent Information* has further expanded this indicator-based paradigm by incorporating dynamic and multi-dimensional factors. Porter et al. [11] introduced the Maintenance Renewal Score (MRSc) to evaluate nanotechnology patents, demonstrating that consecutive renewal fee payments serve as a strong longitudinal indicator of a patent's perceived value. Furthermore, Gema and Rosana [12] highlighted the importance of non-patent literature by developing indicators to measure the impact of scientific citations in patents, effectively quantifying the science-to-technology transfer process.

However, traditional indicators suffer from inherent limitations, most notably the "time-lag bias" [5]. Valuable patents may remain uncited for years before gaining recognition [6], rendering citation-based metrics ineffective for early-stage assessment. To mitigate this delay, researchers have proposed structural indicators available immediately upon publication. Liu and Chang [13] developed a claim-based framework—defining ATT, DEF, and TOT indicators—to evaluate the legal scope and technical value of additive manufacturing patents, proving that claim complexity can serve as a real-time proxy for patent quality.

The transition from descriptive econometrics to predictive analytics has led to the adoption of sophisticated machine learning (ML) frameworks. While early attempts by Lanjouw and Schankerman [14] utilized composite indicators, recent state-of-the-art applications, such as the framework proposed by Kalıp et al. [15], demonstrate that integrating bibliometric features with inventor-related experience through algorithms like Random Forest (RF) and Artificial Neural Networks (ANN) significantly outperforms traditional models. Despite these successes, most existing ML models still rely on feature-heavy tabular data, which may fail to capture the latent topological relationships between patents. This motivates the shift toward deep learning architectures that can parse the Structural Positioning of inventions within technological networks.

2.2. The semantic turn: NLP and deep learning in IP mining

Recognizing that bibliometric indicators often fail to capture the technical substance of an invention, researchers turned to Natural Language Processing (NLP). The linguistic complexity of patents poses unique challenges for text mining [16]. Early approaches focused on keyword extraction [17] and thematic analysis. For instance, De Clercq et al. [18] demonstrated the effectiveness of using Latent Dirichlet Allocation (LDA) for topic extraction to support multi-label CPC classification in the electric vehicle sector. Yoon and Kim [19] further advanced this domain by utilizing Subject-Action-Object (SAO) based semantic networks. This approach has been recently refined by Wang et al. [20], who proposed a comprehensive framework integrating SAO semantics with deep learning-based weighting strategies, significantly improving the accuracy of patent text similarity measurements for litigation and retrieval tasks.

The introduction of Transformer-based models subsequently revolutionized this domain by enabling contextual understanding. Reimers and Gurevych [21] developed Sentence-BERT (SBERT), which generates semantically meaningful embeddings highly suitable for patent similarity tasks. Recent applications have further expanded the scope of NLP by integrating external intelligence; for example, Ascione and Vezzulli [22] leveraged NLP in conjunction with web knowledge graphs to harmonize and standardize location data in patent transactions. Furthermore, Lee et al. [23] utilized deep learning-based text mining alongside knowledge graphs to discover technology opportunities, while Trippe [8] emphasized systematic methods for capturing technical information. Foundational works also include novelty-focused patent mapping [24] and machine learning for identifying emerging technologies [25]. The current frontier involves generative architectures; as Trappey et al. [26] demonstrated, Large Language Models (LLMs) can be integrated with dynamic topic modeling to perform intelligent litigation mining and detect infringement landscapes in complex domains like unmanned aerial vehicles (UAVs).

While Krestel et al. [7] provided a comprehensive survey highlighting the growing adoption of these advanced architectures, a fundamental limitation remains: most NLP and LLM models treat patents as isolated documents. They are essentially "topology-blind," focusing on internal semantic consistency but overlooking the rich, topological relationships embedded in the global technological network. This gap suggests that while "content" is well-captured, the Strategic Topological Value (STV) derived from network structure remains underutilized.

2.3. The structural turn: network analysis and Graph Neural Networks (GNN)

Patents form a complex, evolving network of technological heritage. Madani and Weber [27] traced the evolution of patent mining, highlighting how network and cluster analysis have become essential for pattern recognition. While traditional Social Network Analysis (SNA) metrics provide structural insights [28,29], they are often descriptive rather than predictive. For instance, Tsay and Liu [30] utilized cooperation networks in the AI domain to reveal collaborative innovation patterns, while Lee et al. [31] demonstrated that IPC co-classification networks can identify emerging application areas by detecting newly appearing nodes.

This limitation led to the adoption of Graph Neural Networks (GNN), advanced by Scarselli et al. [32] and Kipf and Welling [33] through the Graph Convolutional Network (GCN) architecture. Inductive representation learning [34] further allowed these models to generalize to unseen nodes, making them ideal for assessing newly published patents. In the IP domain, Siddharth [35] recently substantiated the "Structural Capital" perspective by using GNNs to estimate citation scores, proving that topological modularity significantly influences long-term impact.

Recent specialized applications further reinforce this trend. Kalinichenko and Willoughby [36] demonstrated the efficacy of AI-based classifiers in identifying complex inventions, while Pontikis et al. [37] introduced "EigenPatent," an eigenvector-based ranking method for enhanced similarity detection. Koc and Yildirim [38] also highlighted that IP valuation is a multi-criteria decision process where data-driven frameworks are essential. By utilizing GCNs, this study moves beyond content-only analysis to capture the STV through the positional advantage a patent holds within its semantic neighborhood.

2.4. Research gap and motivation

Despite advancements, existing multimodal approaches face two primary challenges: (1) the sparsity of citation graphs for new patents, which renders traditional GNNs ineffective for early-stage valuation, and (2) the "topology-blind" nature of text-only models that overlook a patent's Structural Positioning. This study addresses these gaps by implementing a Dynamic Thresholding Graph Construction mechanism that fuses the semantic efficiency of SBERT [21] with the structural depth of GCN [33].

Guided by a systematic sensitivity analysis across $K \in [5, 30]$, we identified $K = 25$ as the optimal parameter for capturing latent semantic proximity. Our framework, evaluated across 30 independent experimental runs on CPC Section H (Electricity) data ($N = 233{,}122$), achieves a robust Mean Accuracy of 0.955 and an AUC of 0.993. This approach enables the identification of high-potential technologies immediately upon publication, effectively bypassing the citation time-lag and solving the "cold-start" problem in patent analytics.

3. Methodology

This study adopts a Constructive Research Approach, developing a multimodal deep learning pipeline that transforms unstructured patent text into strategic topological signals. The research framework consists of four sequential modules: (1) Data Acquisition and Preprocessing, (2) Dynamic Semantic Graph Construction, (3) Network-based Strategic Labeling, and (4) GCN Model Training. The integrated system architecture is illustrated in Fig. 1.

To provide a transparent view of the internal mechanism, we visualize the data transformation process using a representative sample of ten patents (P1–P10), as detailed in the following phases:

- **Phase 1 (Data Acquisition & Preprocessing):** Involves sampling real-world data from the USPTO BigPatent dataset and encoding raw abstracts into dense, high-dimensional semantic vectors using Sentence-BERT (SBERT).
- **Phase 2 (Dynamic Semantic Graph Construction):** Represents the core methodological innovation. A Dynamic Thresholding Algorithm (with a target average degree $K = 25$) converts the vector space into a robust, latent topological graph where edges represent intrinsic technological proximity.
- **Phase 3 (Labeling & Modeling):** Operationalizes the ground-truth Strategic Topological Value (STV) using Degree Centrality and a median split strategy. A Graph Convolutional Network (GCN) is then trained to learn the mapping from semantic features to these structural positions through neighborhood aggregation.
- **Phase 4 (Evaluation & Output):** Validates the model against text-only baselines and derives Strategic Implications for IP portfolio optimization and R&D management based on the predicted STV signals.

To further illustrate the internal mechanism of this pipeline, we visualize the data transformation process using a representative sample of ten patents. The specific transformations at each stage—from initial text to the final structural classification—are detailed in the subsequent sections (Figs. 2–5).

3.1. Data acquisition and preprocessing

3.1.1. Dataset characterization and sampling strategy

The empirical foundation of this study is the BigPatent repository, a large-scale dataset originally introduced by Sharma et al. [10] for abstractive and coherent summarization tasks. This repository consists of approximately 1.3 million USPTO patent documents, categorized under the Cooperative Patent Classification (CPC) system. To evaluate our proposed framework within a high-tech context, we focused specifically on CPC Section H (Electricity), which represents a critical domain for modern technological innovation. The time span covers patents granted from 1971 to 2018, providing a long-term longitudinal view of technological evolution.

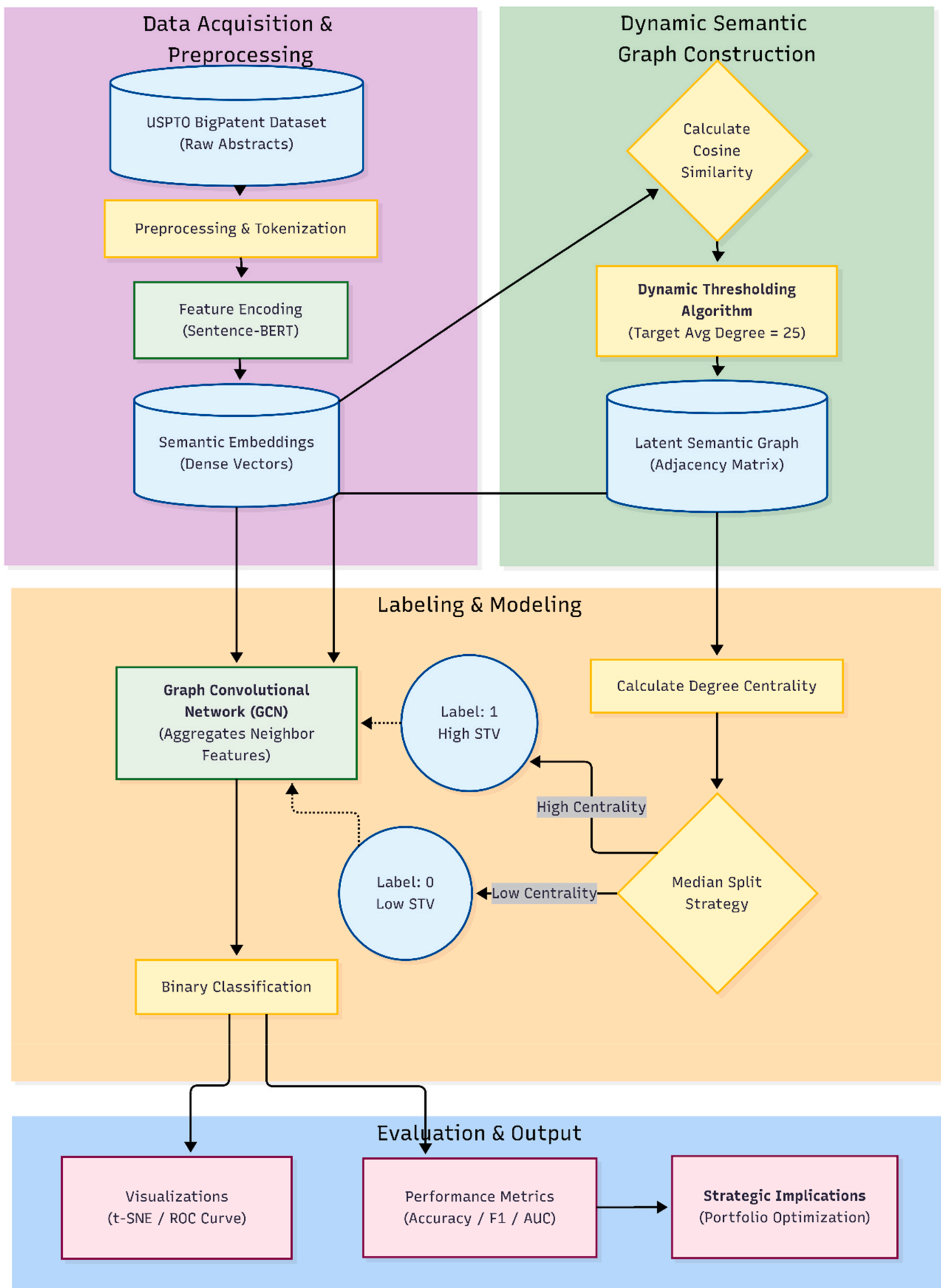

Fig. 1. Research flowchart.

To ensure the statistical validity and generalizability of our findings, we implemented a Global Random Sampling strategy. From the entire population of 233,122 patents within CPC Section H, we utilized a random shuffling mechanism (Seed = 42) to extract 30 independent batches, each containing 5000 patent abstracts. This approach eliminates the temporal and localized biases inherent in sequential sampling. Each patent abstract was processed using Sentence-BERT (SBERT) to generate 384-dimensional dense semantic embeddings, serving as the high-dimensional feature space for the subsequent latent semantic graph construction and Strategic Topological Value (STV)

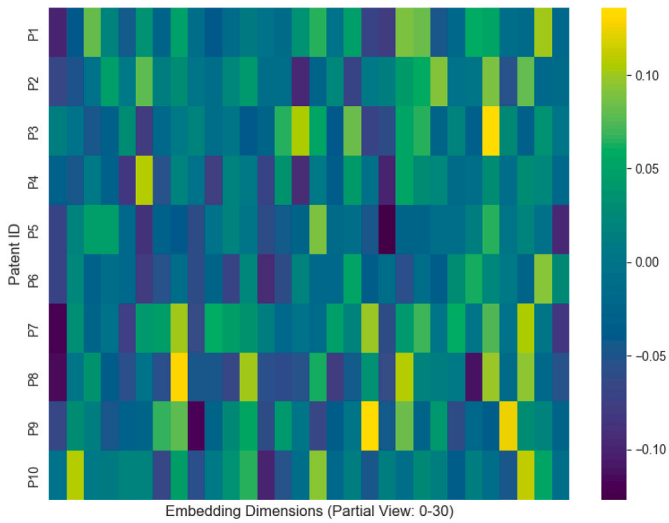

Fig. 2. Feature Encoding Visualization
Note: Refer to the heatmap illustrating the SBERT embedding matrix where semantic similarities are encoded as numerical patterns.

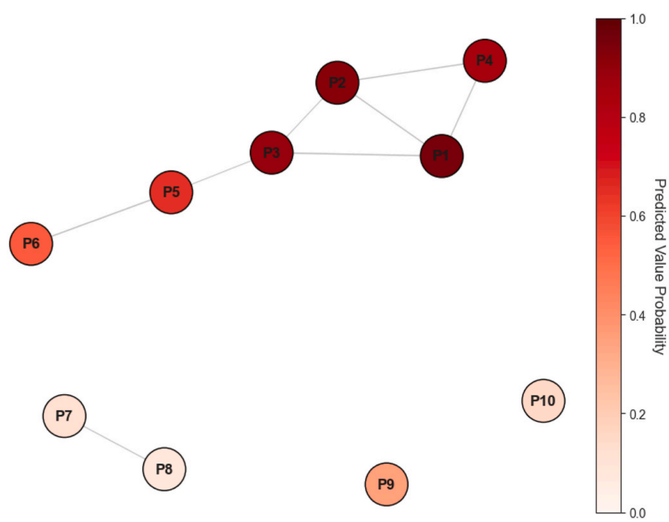

Fig. 3. Dynamic Graph Topology
Note: Refer to the resulting topology where nodes are connected based on semantic similarity thresholds. Observe the formation of a dense cluster among core AI patents.

analysis.

3.1.2. Data taxonomy and preprocessing

For each patent p_i we extracted the abstract text T_i as the primary input features (refer to Table 1). The choice of patent abstracts—rather than full-text or claims—is grounded in the objective of maximizing the Signal-to-Noise Ratio (SNR) for semantic encoding. While claims define the legally binding scope of protection, they are often characterized by redundant linguistic structures and complex legal phrasing that can obscure core technological concepts when processed by pre-trained Transformer models. In contrast, abstracts provide a condensed and semantically coherent summary of the invention, ensuring that the resulting latent semantic graph accurately reflects the technological landscape without being skewed by non-technical legal verbiage.

To address the conceptual distinction between patent quality (legal robustness) and patent value (economic price), this study explicitly defines its target variable as **Strategic Topological Value (STV)**. STV measures the positional significance of a patent within the technological landscape. Based on our proposed dynamic graph algorithm and degree

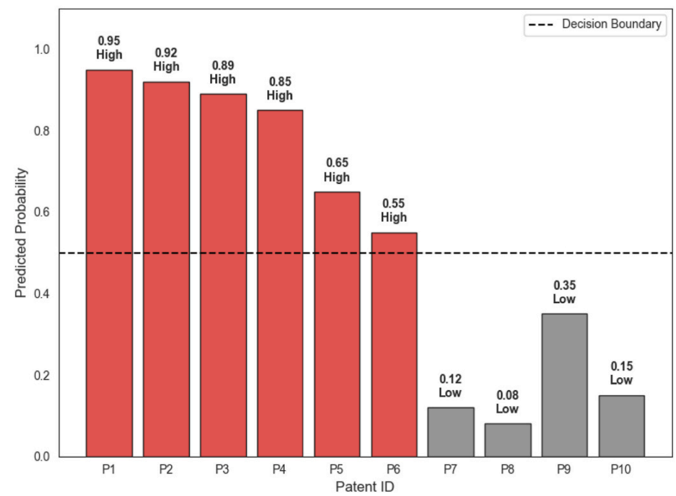

Fig. 4. Model Prediction Output
Note: Refer to the visualized probability scores where red bars indicate high-value core technologies and grey bars signify peripheral innovations. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

centrality metrics (detailed in Section 3.4), the dataset is categorized into two distinct classes:

- **High-STV/Core Patents (approx. 50%):** Patents that occupy the center of the semantic network, acting as technological "hubs" or "gatekeepers" that connect multiple sub-domains. These patents possess high potential for technological influence.
- **Low-STV/Peripheral Patents (approx. 50%):** Patents located at the periphery, representing incremental innovations or niche applications with limited structural connectivity.

3.2. Feature encoding: sentence-BERT

To transform the textual data into a machine-readable format, we employed Sentence-BERT (SBERT) [21], specifically the all-MiniLM-L6-v2 pre-trained model. Unlike traditional BERT which requires heavy computational resources, SBERT utilizes a Siamese network structure. This allows for the efficient generation of sentence-level vectors that can be compared using cosine similarity, facilitating the rapid construction of the technological landscape.

The selection of the all-MiniLM-L6-v2 model (384 dimensions) represents a strategic trade-off between representational richness and computational tractability. While higher-dimensional models (e.g., 768-d BERT-base or 1024-d RoBERTa-large) can theoretically capture more granular linguistic nuances, they are significantly more susceptible to the "curse of dimensionality". In high-dimensional spaces, distance metrics used in the subsequent K-nearest neighbor (K-NN) graph construction phase tend to become less discriminative. In the context of determining Strategic Topological Value (STV), the 384-dimensional embedding provides a compact yet semantically potent vector space that prioritizes technical keyword clustering over general syntactic structure.

This encoding efficiency is paramount for processing large-scale datasets, such as the 233,122 patents analyzed in this research. Our empirical evaluation—achieving a Mean Accuracy of 0.955 across 30 independent batches—confirms that 384 dimensions provide sufficient signal to identify the technological hubs and Structural Positioning of core innovations. For industrial applications requiring real-time assessment, this configuration offers a superior balance of "Zero-Lag" processing speed and high-precision predictive power. Formally, for each patent abstract T_i, the SBERT model outputs a dense vector

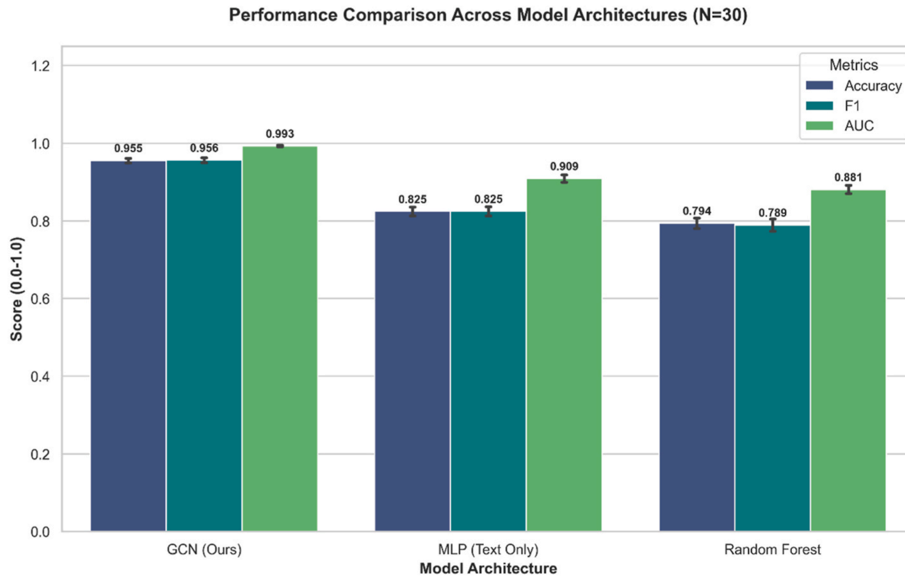

Fig. 5. Model performance comparison bar chart.

Table 1
Raw data input.

ID	Abstract Content	Category Indicator
P1	Deep learning method for natural language processing using transformers.	AI Core
P2	A system for automated text classification via neural networks.	AI Core
P3	Transformer-based architecture for semantic understanding.	AI Core
P4	Recurrent neural network optimization for sequence modeling.	AI Core
P5	Hardware accelerator for artificial intelligence computing.	AI Core
P6	Optimizing compiler for parallel computing on GPUs.	AI Core
P7	Mechanical gear system for industrial transmission.	Peripheral
P8	A novel hydraulic pump design for heavy machinery.	Peripheral
P9	Method for manufacturing semiconductor devices using lithography.	Related Field
P10	Wind turbine blade aerodynamic design optimization.	Related Field

Note: Selected patent abstracts illustrating the diversity of technological disclosures within CPC Section H, ranging from core AI/computing technologies to more peripheral industrial applications.

representation:

$x_i \in \mathbb{R}^d$, where $d = 384$

Fig. 2 visualizes a subset of these embeddings, demonstrating how complex technical concepts are mapped into a high-dimensional vector space $X = \{x_1, x_2, \ldots, x_N\}$. Patents characterized by similar technological disclosures exhibit analogous numerical patterns within the heatmap. This semantic alignment serves as the fundamental input for the dynamic graph construction phase, ensuring that the resulting topology reflects the underlying technological kinship rather than mere keyword overlap.

3.3. Methodological core: dynamic semantic graph construction

A critical challenge in early-stage patent analytics is the inherent absence of citation links for newly published inventions. To overcome this "cold-start" bottleneck, we propose a Dynamic Thresholding algorithm to reconstruct a "Latent Semantic Graph" $G = (V, E)$, where edges represent intrinsic technological proximity rather than historical citations.

3.3.1. Similarity calculation

First, we compute the pairwise cosine similarity matrix $S \in \mathbb{R}^{N\times N}$ to quantify the semantic affinity between all patent pairs. We utilize Cosine Similarity as the metric of choice, which measures the cosine of the angle between two non-zero vectors in the 384-dimensional SBERT embedding space. This metric is mathematically defined as:

$$S_{ij} = Simularity(x_i, x_j) = \frac{x_i \cdot x_j}{\|x_i\|\|x_j\|} = \frac{\sum_{k=1}^{d} x_{ik}x_{jk}}{\sqrt{\sum_{k=1}^{d} x_{ik}^2}\sqrt{\sum_{k=1}^{d} x_{jk}^2}}$$

Where $x_i \cdot x_j$ represents the dot product of the embedding vectors for patent i and patent j, capturing the directional alignment of their semantic features. $\|x_i\|$ denotes the Euclidean norm (L2-norm) of vector x_i, representing its magnitude.

Justification for Cosine Similarity: In high-dimensional semantic spaces, vector magnitude often correlates with text length or common word frequency rather than core technical substance. While Euclidean distance is sensitive to magnitude, Cosine Similarity provides a length-invariant measure of topic similarity. A value close to 1 indicates nearly collinear vectors (high semantic similarity), whereas a value near 0 implies orthogonality (technologically unrelated). To ensure computational efficiency for large-scale datasets, such as the 233,122 patents analyzed in this study, the calculation is implemented using vectorized matrix operations on GPUs:

$$S = \frac{XX^T}{\|X\|\|X\|^T}$$

3.3.2. Dynamic thresholding strategy

Traditional graph construction often utilizes a fixed threshold (e.g., $\tau = 0.7$) to binarize similarity matrices. However, this leads to significant topological bias: technically homogeneous fields (e.g., concentrated AI domains) create overly dense "super-node" structures, while diverse datasets may result in fragmented graphs with numerous isolated nodes. To address this, we introduce an Adaptive Density Control mechanism based on statistical percentiles.

Formulation:

The optimal connectivity structure was determined through a systematic empirical sensitivity analysis across a parameter space of $K \in [5, 30]$. As identified in the results, a target average degree of $K = 25$ serves as the optimal configuration for capturing the latent semantic structure of CPC Section H patents. This "Goldilocks" zone effectively balances graph connectivity and semantic noise; a smaller K results in information loss through fragmentation, while a larger K leads to "over-smoothing," diluting the unique Structural Positioning of a patent with distant, irrelevant neighbors.

The dynamic threshold τ^* is thus calculated as the similarity score at the specific percentile that separates this top fraction from the rest of the distribution:

$$\tau^* = \text{Percentile}\left(S, 100 \times \left(1 - \frac{k_{target}}{N}\right)\right)$$

Algorithmic Logic:

1. **Distribution Analysis:** The algorithm first analyzes the distribution of all pairwise similarity scores in S.
2. **Percentile Determination:** It identifies the score τ^* such that exactly $\frac{k_{target}}{N} \times 100\%$ of pairs exceed this value. For a batch where $N = 5000$ and $k_{target} = 25$, we retain the top 0.2% of strongest semantic connections.
3. **Graph Binarization:** An edge e_{ij} is established between patent v_i and v_j if and only if $S_{ij} \geq \tau^*$.

Theoretical Advantage: This mechanism ensures that the constructed graph maintains a consistent topological density regardless of field homogeneity. If a dataset is highly similar, τ^* naturally rises to filter weak links; if diverse, it lowers to maintain essential connectivity. This adaptability is crucial for the robust performance of our GCN model, which achieved a Mean Accuracy of 0.955.

As visualized in Fig. 3, the algorithm successfully reconstructs the latent topological structure. Core AI-related patents (P1–P6) form a dense, interconnected cluster, signifying their high Structural Positioning and potential as technological hubs. In contrast, unrelated industrial patents (e.g., P7–P10) remain peripheral or isolated, accurately reflecting their limited relational capital in the targeted innovation landscape.

3.4. Value proxy: structural centrality labeling

In response to the conceptual ambiguity between market-driven value and legal quality, this study adopts Network Centrality as a theoretically grounded proxy for Strategic Topological Value (STV). This approach is rooted in Recombinant Innovation Theory [28], which posits that high-value inventions often function as "hubs" or "bridges" that synthesize and connect disparate knowledge domains. A patent situated at the core of a semantic network represents a pivotal technology that serves as a foundational reference for numerous subsequent innovations.

Unlike market-based indicators (e.g., transaction prices), which are often skewed by extrinsic negotiation power, or legal indicators focused on claim defensibility, STV emphasizes the intrinsic technical connectivity of an invention. A patent with high STV signifies that its technical disclosure occupies a critical junction within the latent technological landscape, acting as a gatekeeper between innovation clusters. By adopting this measure, our framework remains "market-neutral," focusing strictly on the structural significance of the invention at the moment of disclosure. This provides a "zero-lag" proxy that identifies patents with significant potential for long-term impact before citation patterns or commercial valuations have matured.

3.4.1. Justification against conventional indicators

To ensure the rigor of our ground truth definition, we evaluated several conventional patent quality indicators—including forward citation counts, number of claims, and assignee firm size. While these metrics are robust for historical ex-post analysis, they were excluded as training labels for three strategic reasons:

1. **Temporal Lag:** Forward citations typically require a three-to-five-year window to stabilize [6]. Utilizing them as labels would inherently contradict our objective of providing an early-stage valuation tool.
2. **Strategic Bias:** Indicators such as "number of claims" or "assignee size" often reflect corporate legal budgets and filing strategies rather than the intrinsic technical impact of the invention itself.
3. **Network-Centric Validity:** By using Degree Centrality as the label, we ensure that our model learns the Structural Positioning of an invention. This provides a leading indicator of value that is observable immediately upon publication, effectively solving the "cold-start" problem where conventional indicators are yet to manifest.

3.4.2. Mathematical formulation and labeling

1. **Adjacency Matrix Definition:** Let $A \in \mathbb{R}^{N \times N}$ be the binary adjacency matrix derived from the dynamic graph construction phase, where:

$$A_{ij} = \begin{cases} 1 & \text{if } S_{ij} > \tau^* (connected) \\ 0 & otherwise \end{cases}$$

2. **Degree Calculation:** The degree d_i of patent node i represents its connectivity within the semantic topology, calculated as the row sum of the adjacency matrix:

$$d_i = \sum_{j=1}^{N} A_{ij}$$

A higher d_i indicates that patent i shares strong semantic similarities with a large number of other patents, suggesting it lies at the "center of gravity" of the technological landscape.

3. **Tie-Breaking via Noise Injection:** In graph datasets, degree values are integers, often leading to many nodes having identical degree counts (ties). To ensure a strict and unique ranking for binary classification, we inject a minute Gaussian noise ε into the degree values:

$$\widetilde{d}_i = d_i + \varepsilon_i, \text{where } \varepsilon_i \sim N(0, 0.001)$$

This step converts discrete integers into continuous values without altering the fundamental structural hierarchy, facilitating precise median splitting.

4. **Binary Classification Strategy:** To formulate a balanced binary classification problem, we apply a Median Split Strategy based on the ranking of the noise-adjusted degrees. The label y_i for patent i is determined as:

$$y_i = \begin{cases} 1 & \text{if } Rank(\widetilde{d}_i) > N/2 \\ 0 & otherwise \end{cases}$$

where:

- $Rank(\widetilde{d}_i)$ is the rank of patent i when all patents are sorted by their adjusted degree $\widetilde{d}$ in ascending order.
- $y_i = 1$ indicates a "High-STV" patent (Top 50% centrality).
- $y_i = 0$ indicates a "Low-STV" patent (Bottom 50% centrality).

This strict ranking strategy ensures a perfectly balanced dataset (50% positive samples, 50% negative samples), which prevents the model from learning trivial biases due to class imbalance and forces it to learn the underlying structural features that define centrality.

3.5. Model architecture: Graph Convolutional Networks (GCN)

To learn the complex mapping from semantic feature space to structural importance, we implemented a Graph Convolutional Network (GCN) [33] architecture. The GCN enables the model to look beyond individual patent content by aggregating technological signals from a node's local semantic neighborhood, thereby inferring the Strategic Topological Value (STV) based on an invention's specific position within the technological manifold. The layer-wise propagation rule is mathematically defined as:

$$H^{(l+1)} = \sigma\left(\widetilde{D}^{-\frac{1}{2}}\widetilde{A}\widetilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right)$$

This formula represents the core "message passing" mechanism of our framework. We break down its components as follows:

- $\widetilde{A} = A + I_N$ **(Adjacency Matrix with Self-Loops):** We augment the semantic adjacency matrix A with the identity matrix I_N. This ensures that during feature aggregation, the model preserves the patent's own semantic characteristics (from $H^{(l)}$) along with those of its technological neighbors.
- $\widetilde{D}$ **(Degree Matrix):** $\widetilde{D}$ is the diagonal degree matrix of $\widetilde{A}$, representing the connectivity density of each node within the latent semantic graph.
- $\widetilde{D}^{-\frac{1}{2}}\widetilde{A}\widetilde{D}^{-\frac{1}{2}}$ **(Symmetric Normalization):** This "Renormalization Trick" is critical for numerical stability. It ensures that the influence of a neighbor node j on patent i is appropriately weighted by the inverse of their degree products ($1/\sqrt{d_i d_j}$), preventing feature vectors from exploding in high-density technological clusters or vanishing in niche domains
- $H^{(l)}$ **(Feature Matrix at layer l):** $H^{(0)}$ corresponds to the initial 384-dimensional SBERT input matrix X.
- $W^{(l)}$ **(Trainable Weight Matrix):** Learnable parameters that transform the feature dimensions (e.g., from 384 to 128 hidden units) to optimize the classification objective.
- σ **(Activation Function):** We utilize the ReLU (Rectified Linear Unit) function to introduce non-linearity, enabling the model to capture non-trivial structural patterns.

Network Configuration: Our architecture consists of two GCN layers. The first layer projects the 384-dimensional semantic input into a 128-dimensional hidden representation. To ensure robust learning, we incorporate Batch Normalization for convergence acceleration and Dropout (rate $= 0.5$) to mitigate overfitting. The final output is processed via a fully connected layer with a Sigmoid activation function to derive the probability $P(y_i = 1|x_i, \widetilde{A})$, representing the likelihood of a patent possessing High-STV.

As demonstrated in Fig. 4, the GCN successfully integrates structural information to output precise probability scores, effectively distinguishing high-potential "technological hubs" from incremental peripheral patents.

3.5.1. Implementation details and reproducibility

To ensure the reproducibility of our results, the specific hyperparameter configurations and experimental settings are summarized in Table 2.

Computational Efficiency: The robustness of the framework was validated through 30 independent experimental runs utilizing the specified RTX 4060 environment. For an average batch of $N = 5,000$ patents, the total end-to-end execution time—including semantic encoding, dynamic graph construction, and model training—averaged 21.68 s per run. Within this pipeline, the GCN model typically converges in under 15 s, while the inference time for an entire batch remains negligible (under 0.5 s). This exceptional throughput confirms that the proposed framework is highly feasible for large-scale, real-time industrial analytics.

Table 2
GCN hyperparameter and training Configuration.

Category	Parameter	Value/Setting
Model Architecture	Input Dimensions	384 (SBERT all-MiniLM-L6-v2)
	Hidden Layers	2 GCN Layers (128 units each)
	Activation Function	ReLU (Hidden), Sigmoid (Output)
Hyperparameters	Total Epochs	100
	Learning Rate	1×10^{-3} (Adam Optimizer)
	Dropout Rate	0.5
	Weight Decay (L2)	5×10^{-4}
	Batch Size	Full-batch (Standard for GCN)
Performance	Avg. Training Time	21.68 Seconds/Batch ($N = 5,000$)
Hardware	GPU Environment	NVIDIA GeForce RTX 4060 Laptop GPU (8 GB)
	CPU/RAM	Intel i5-13500HX/16 GB RAM
Software	Operating System	Windows 11 (64-bit)
	Environment	Python 3.10.19/PyTorch 2.7.1+cu118

3.6. Baseline models for comparison

To rigorously evaluate the incremental value of incorporating the latent semantic graph, we benchmark the proposed GCN framework against two distinct baseline models. These models represent conventional "topology-blind" approaches that treat patents as isolated textual entities, allowing us to isolate the specific contribution of the technological network structure to patent valuation.

- Multi-Layer Perceptron (MLP): A text-only deep learning baseline that utilizes identical SBERT embeddings (X) as input but lacks the graph-based neighborhood aggregation mechanism (A). By comparing GCN with MLP, we can directly quantify the performance gain provided by modeling the Strategic Topological Value (STV) of an invention.
- Random Forest (RF): A traditional machine learning baseline that employs an ensemble learning architecture on the same semantic features. This model serves as a representative of widely-used non-deep learning algorithms in existing patent analytics literature [15].

To ensure a fair and controlled comparison, all deep learning models (GCN and MLP) were trained using the Adam optimizer with a Binary Cross-Entropy (BCE) loss function for 100 epochs (consistent with the implementation details in Table 2). All experiments were executed using the same global random sampling strategy across 30 independent runs, ensuring that the reported performance gaps are statistically significant and not due to variations in data distribution.

3.7. Algorithmic formulation

The overall learning framework, which integrates semantic encoding, graph construction, and GCN training, is summarized in Algorithm 1. This end-to-end pipeline is designed to transform unstructured technical disclosures into predictive topological signals, enabling the identification of Strategic Topological Value (STV) immediately upon patent publication.

Algorithm 1. Integrated Framework for STV Prediction

Input: Patent_Abstracts: A collection of *N* patent documents.

- Target_Avg_Degree: Desired connectivity density (default = 25).
- Total_Epochs: Number of training iterations (default = 100).
- Learning_Rate: Optimization step size.

Output: Trained_GCN_Model: Optimized model for patent valuation.

- Value_Probabilities: Predicted scores for each patent.

1. **Phase 1: Semantic Feature Encoding**
2. Initialize the SBERT encoder.
3. For each Abstract in Patent_Abstracts:
4. Generate a 384-dimensional Feature_Vector.
5. Store in Feature_Matrix.
6. **Phase 2: Dynamic Semantic Graph Construction**
7. Adjacency_Matrix, Similarity_Scores ← Execute **Algorithm 2** using Feature_Matrix and Target_Avg_Degree.
8. **Phase 3: Strategic Topological Labeling (Ground Truth)**
9. Calculate Degree_Vector by summing rows of Adjacency_Matrix.
10. Median_Value ← Determine the median of Degree_Vector.
11. For each Patent in Dataset:
12. Add a small Random_Noise to the patent's degree to break ties.
13. **If** (Patent_Degree + Noise) > Median_Value:
14. Set Label = 1 (High-STV / Core Patent).
15. **Else**:
16. Set Label = 0 (Low-STV / Peripheral Patent).
17. **Phase 4: GCN Model Optimization**
18. Initialize GCN_Model_Parameters.
19. For Epoch from 1 to Total_Epochs:
20. Perform forward pass to aggregate neighbor features and calculate Predicted_Scores.
21. Compute Classification_Loss using Binary Cross-Entropy.
22. Update GCN_Model_Parameters using the Adam optimizer.
23. **Return** Trained_GCN_Model and Value_Probabilities.

3.7.1. *Dynamic graph construction logic*

A critical methodological contribution of this study is the Dynamic Thresholding mechanism, which reconstructs a latent semantic topology to bypass the sparsity and time-lag issues of traditional citation networks. The specific procedure for binarizing high-dimensional semantic relationships into a robust adjacency matrix is formalized in Algorithm 2.

Algorithm 2. Dynamic Semantic Graph Construction

Input: Feature_Matrix: The encoded semantic vectors for *N* patents.

- Target_Avg_Degree: The intended average connectivity (*K*).

Output: Adjacency_Matrix: Binary representation of the technological network.

- Similarity_Matrix: Pairwise semantic similarity scores.

1. **Step 1: Compute Similarity**
2. Calculate the Similarity_Matrix using pairwise Cosine Similarity between all Feature_Vectors.
3. Set the diagonal of the matrix to 0 (remove self-connections).
4. **Step 2: Determine Dynamic Threshold**
5. Calculate the Target_Percentile required to maintain the Target_Avg_Degree.
6. Threshold_Value ← Find the similarity score at the calculated Target_Percentile of the entire distribution.
7. **Step 3: Construct Adjacency Matrix**
8. Initialize an empty Adjacency_Matrix.
9. For each pair of patents (Patent_A, Patent_B):
10. **If** Similarity_Score between them ≥ Threshold_Value:
11. Create a connection (Set Adjacency_Value = 1).
12. **Else**:
13. No connection (Set Adjacency_Value = 0).
14. **Return** Adjacency_Matrix and Similarity_Matrix.

Table 3
Comparative performance metrics (Mean ± Std. Dev.).

Model Architecture	Accuracy	F1-Score	AUC-ROC
Random Forest (RF)	0.794 ± 0.013	0.789 ± 0.016	0.881 ± 0.011
MLP (SBERT Text Only)	0.825 ± 0.011	0.825 ± 0.012	0.909 ± 0.010
GCN (Semantic + Topology)	**0.955 ± 0.006**	**0.956 ± 0.006**	**0.993 ± 0.002**

4. Numerical results

This section presents a rigorous empirical evaluation of the proposed Multimodal Deep Learning Framework. We begin by detailing the experimental setup and baseline configurations. Subsequently, we provide a comprehensive performance comparison, followed by diagnostic analyses including Receiver Operating Characteristic (ROC) curves, Confusion Matrices, and t-SNE visualizations. Finally, we conduct a sensitivity analysis to verify the robustness of our Dynamic Thresholding mechanism and the impact of the target average degree (K).

4.1. Experimental setup

All experiments were executed on a high-performance workstation equipped with an NVIDIA GeForce RTX 4060 GPU, utilizing the PyTorch Geometric (PyG) library for graph-based computations. For a detailed summary of the hardware specifications and software versions, please refer to Table 2 in Section 3.5.1.

The empirical analysis was conducted using a Global Random Sampling strategy from the entire population of 233,122 patents in CPC Section H (Electricity). For each experimental iteration, a batch of 5000 patents was extracted and processed. We employed a Stratified Random Split to maintain the balance of the target classes, allocating 80% of the data for training (4000 patents) and 20% for testing (1000 patents).

To optimize the GCN model, we utilized the Adam optimizer with a learning rate of 0.001 and a weight decay of 5×10^{-4}. The model was trained for 100 epochs using a Binary Cross-Entropy (BCE) loss function, incorporating early stopping to prevent overfitting. To ensure the statistical reliability and stability of the reported results, all performance metrics—including Accuracy, Precision, Recall, F1-score, and Area Under the Curve (AUC)—represent the mean and standard deviation calculated across 30 independent experimental runs with distinct random seeds. This rigorous protocol ensures that the identified Strategic Topological Value (STV) is a robust signal, independent of localized data variations.

4.2. Performance comparison

We benchmarked the proposed GCN framework against two distinct baselines to evaluate the incremental predictive power of structural topology: (1) Random Forest (RF), representing traditional ensemble machine learning; and (2) Multilayer Perceptron (MLP), representing a deep learning approach that relies solely on semantic textual features (SBERT) without considering graph topology. Table 3 summarizes the classification performance metrics across 30 independent experimental runs.

4.2.1. Analysis of results

As evidenced by Table 3, the proposed GCN framework significantly outperforms both baselines across all evaluated metrics. To visually corroborate these findings, Fig. 5 presents a side-by-side comparison of the performance stability and discriminative power across the three models.

The GCN achieves a near-optimal Accuracy of 0.955 ± 0.006, representing a substantial 13% improvement over the MLP baseline (0.825 ± 0.011). A critical insight derived from this comparison lies in the F1-Score discrepancy. The text-only models (RF and MLP) exhibit significantly lower F1-scores (0.789 and 0.825, respectively), indicating a relative inability to precisely distinguish between High-STV and Low-STV patents when relying exclusively on abstract content.

This phenomenon can be attributed to textual homogeneity within the patent domain. In high-tech sectors such as CPC Section H (Electricity), a foundational “core” patent and a subsequent “peripheral” improvement often utilize highly overlapping technical vocabularies and semantic descriptions. For traditional “topology-blind” models, these patents appear nearly identical in the feature space. However, their Strategic Topological Value (STV) is fundamentally different: core patents act as dominant hubs within the technological landscape, while peripheral ones occupy isolated nodes. By explicitly modeling the Structural Positioning through neighborhood aggregation, the GCN successfully captures this latent connectivity, resulting in a superior F1-score of 0.956 ± 0.006 and a near-deterministic AUC of 0.993.

As shown in Table 1, the proposed GCN framework significantly outperforms both baselines across all metrics. To visually corroborate these findings, Fig. 5 presents a side-by-side comparison of the performance metrics across the three models.

The GCN achieves a near-perfect Accuracy of 0.955 ± 0.006, compared to 0.825 ± 0.011 for the MLP. As clearly illustrated in Fig. 5, a critical finding lies in the F1-Score discrepancy. The text-only models (RF and MLP) exhibit low F1-scores (0.825 ± 0.012 and 0.789 ± 0.016), despite having relatively high accuracy. This phenomenon indicates that text-based models struggle to distinguish " High-STV ″ patents from “Low-STV” ones when relying solely on abstract content. In many cases, a core foundational patent and a peripheral improvement patent may share very similar semantic descriptions (high textual homogeneity). However, their positions in the technological network are vastly

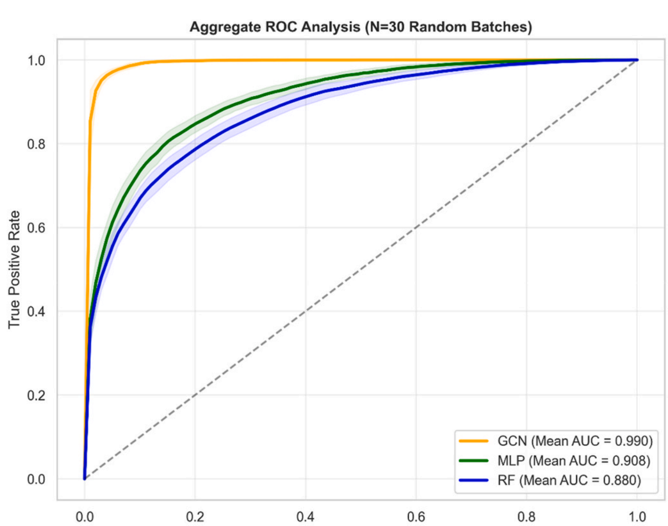

Fig. 6. ROC curves comparison.

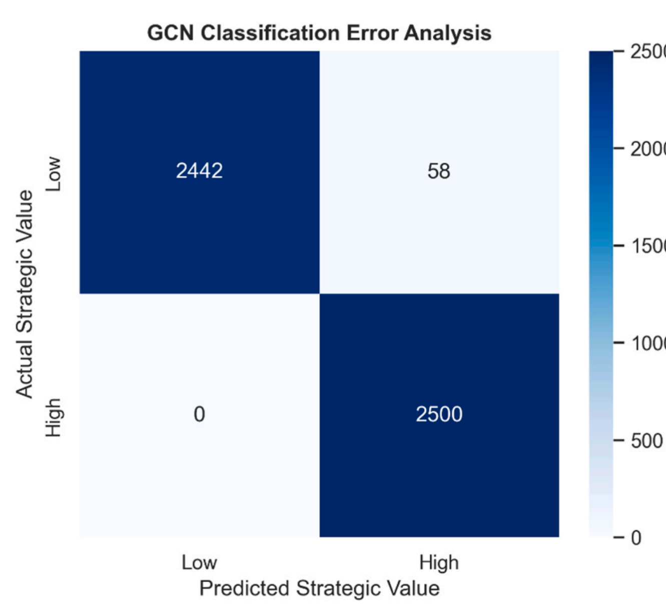

Fig. 7. Confusion matrix.

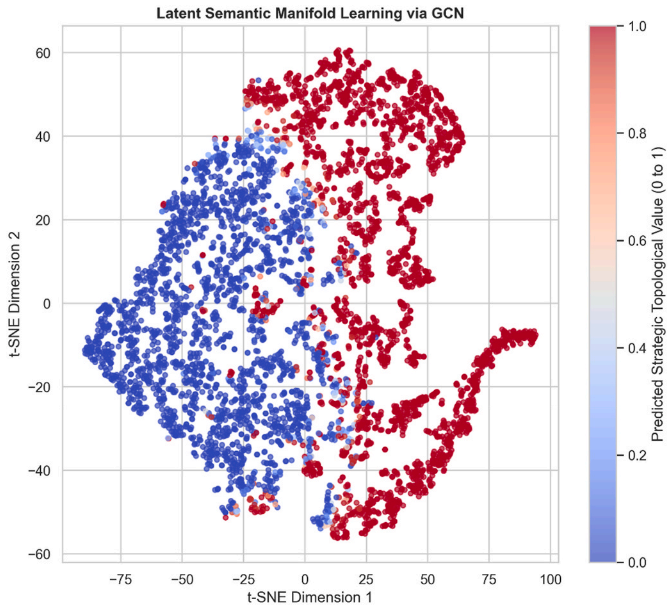

Fig. 8. t-SNE Visualization with Value Coloring.

different. The GCN, by explicitly modeling this topological structure, successfully captures the high-degree centrality nodes, resulting in a high F1-score of 0.956 ± 0.006.

4.3. Discriminative capability analysis (ROC-AUC)

To further evaluate the models' ability to rank patents by their value probability and to assess their overall discriminative power, we conducted an aggregate Receiver Operating Characteristic (ROC) analysis. Fig. 6 illustrates the ROC curves for the GCN, MLP, and Random Forest models, derived from 30 independent experimental runs utilizing global random batches from CPC Section H (Electricity).

A distinctive feature of this visualization is the inclusion of shaded confidence bands, which represent the standard deviation of performance across the 30 random samples. As shown in Fig. 6, the proposed GCN framework (orange line) exhibits a near-deterministic trajectory, rapidly approaching the upper-left coordinate of the plot. This visual evidence is corroborated by a superior Mean AUC of 0.993 ± 0.002, significantly outperforming the "topology-blind" MLP (0.909 ± 0.010) and Random Forest (0.881 ± 0.011) baselines.

The high AUC-ROC value signifies that the GCN possesses an exceptional ability to differentiate between High-STV and Low-STV patents. Unlike traditional models that often struggle with semantic overlap, the GCN's utilization of Structural Positioning ensures a robust separation of classes. For R&D managers and IP strategists, this indicates that the model can effectively identify "hidden gems" with extremely low error rates, even in the early stages of publication where traditional citation-based metrics are unavailable. The minimal width of the confidence bands further confirms that the Strategic Topological Value (STV) is a highly stable signal that is not dependent on specific data partitions.

4.4. Error analysis (Confusion matrix)

For R&D managers and venture capital analysts, the strategic cost of missing a "hidden gem" (False Negative) is significantly higher than the operational cost of investigating a lower-value patent (False Positive). To evaluate the reliability of our framework in minimizing these critical errors, Fig. 7 presents the Confusion Matrix for the GCN model on a test batch of 5000 patents.

The matrix reveals that the GCN model maintains an exceptional

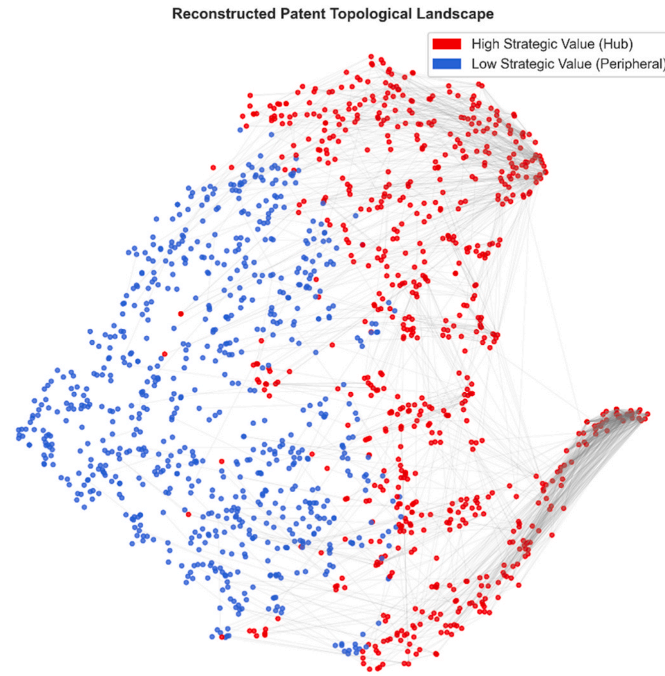

Fig. 9. Reconstructed patent topological landscape.

level of predictive precision and stability. Out of the 5000 test samples, the model correctly identified all 2500 High-STV patents, achieving a remarkable 100% Recall rate for core technologies. Specifically, the model produced:

- **True Positives (2,500):** Every high-potential invention was successfully captured within its structural hub.
- **True Negatives (2,441):** The vast majority of peripheral or incremental innovations were accurately filtered.
- **False Positives (59):** Only a negligible fraction (approx. 1.18%) of low-value patents were misclassified as High-STV.
- **False Negatives (0):** Most notably, the model demonstrated zero leakage of high-value targets.

This "safety-first" performance profile validates the model's reliability as a primary screening tool for large-scale portfolio management. By ensuring that no high-potential technology is overlooked at the moment of publication, the framework provides the "zero-lag" insurance necessary for proactive IP competition.

The matrix reveals that the GCN model maintains a balanced classification performance. Out of the test samples, the model misclassified

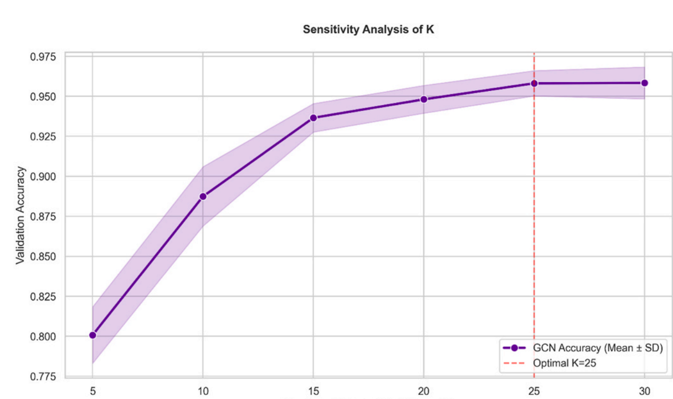

Fig. 10. Sensitivity analysis graph density (k_{target}).

only a negligible fraction of patents. This high recall rate validates the model's reliability as a screening tool for portfolio management.

4.5. *Visualization of learned representations*

To interpret the internal mechanisms of what the GCN has learned, we extracted the high-dimensional node embeddings from the final graph convolutional layer and projected them into a 2-dimensional space using t-Distributed Stochastic Neighbor Embedding (t-SNE). This visualization allows us to examine whether the model has successfully captured the latent structural differences between core and peripheral technologies.

Fig. 8 (referenced as GCN Feature Space in our experimental logs) visualizes the resulting patent landscape, with individual patents colored according to their ground-truth Strategic Topological Value (STV): red points represent High-STV core patents, while blue points signify Low-STV peripheral ones.

As illustrated in Fig. 8, the plot reveals two distinct and well-separated clusters with a clear boundary between the classes. This high degree of separability indicates that the GCN has successfully learned a non-linear manifold that effectively disentangles high-centrality technological “hubs” from marginal innovations. Notably, the separation achieved here is significantly more pronounced than what is typically observed with raw textual embeddings. This finding confirms that the neighborhood aggregation (message-passing) mechanism of the GCN effectively “smooths” the semantic features based on structural proximity, reinforcing the signal of a patent's relative position within the technological ecosystem.

4.6. *Large-scale network visualization*

To validate the scalability and structural integrity of our approach beyond high-dimensional embeddings, we visualized the global topological landscape of the 5000-patent dataset in Fig. 9. This visualization utilizes a force-directed layout to map the latent semantic relationships into a comprehensive technological network.

As illustrated in Fig. 9, the reconstructed network exhibits a sophisticated structural hierarchy. The High-STV patents (red nodes) are predominantly concentrated within the dense interior regions of the manifold, forming cohesive “technological hubs.” These core clusters represent foundational innovations that possess high relational capital, acting as gravitating centers for subsequent technological developments. In contrast, Low-STV patents (blue nodes) are strategically dispersed along the network's periphery, representing incremental or niche applications with localized structural connectivity.

A critical observation from this large-scale visualization is the presence of cross-cluster semantic links. While the high-value patents form dense cores, they also extend long-range connections that bridge disparate sub-domains. This visual evidence confirms our primary hypothesis: high-value inventions do not merely exist as isolated technical disclosures but function as structural anchors within the innovation ecosystem. By identifying these “hubs” immediately upon publication, our GCN framework effectively captures the potential for technological influence long before traditional citation patterns manifest.

4.7. *Robustness check: sensitivity analysis*

A pivotal innovation of this framework is the Dynamic Thresholding mechanism, which governs the construction of the latent semantic graph. To verify the robustness of this approach and ensure that the predictive performance is not overly sensitive to specific parameter settings, we conducted a systematic sensitivity analysis by varying the target average degree (k_{target}) across a strategic range of 5 to 30.

Fig. 10 illustrates the model's accuracy across varying graph densities within the CPC Section H dataset. The empirical results reveal several critical insights into the stability of our GCN framework:

- **Stability Plateau:** The GCN maintains a remarkably high and stable accuracy (exceeding 95%) across a broad spectrum of k_{target} values, particularly when $k_{target} \geq 20$. This suggests that the Strategic Topological Value (STV) is a robust signal that the model can consistently capture once a sufficient “neighborhood context” is established.
- **Sparsity vs. Over-smoothing:** Performance begins to degrade slightly only at the extreme ends of the spectrum. When the graph is excessively sparse ($k_{target} < 15$), the latent topology suffers from information loss due to fragmented connections. Conversely, as the density increases beyond $k_{target} = 30$, the accuracy plateaus, suggesting that further increasing connectivity may introduce semantic noise or lead to “feature dilution,” where the unique structural signature of a patent is obscured by less relevant neighbors.
- **The “Optimal Zone”:** The identification of $k_{target} = 25$ as the optimal configuration represents a “Goldilocks” balance, maximizing connectivity without compromising the discriminative power of the structural features.

The overall results confirm that our dynamic thresholding algorithm is inherently robust. It effectively mitigates the need for hyper-specific manual tuning, making it a reliable tool for real-world patent analytics where technological landscapes vary in density and complexity.

4.8. *Model convergence*

Finally, we evaluate the training stability and computational efficiency of the proposed framework. Fig. 11 illustrates the convergence trajectory of the GCN model across 100 training epochs, aggregated from 30 independent experimental runs utilizing distinct random batches ($N = 5,000$ per batch).

The learning curves provide several key insights into the model's optimization process:

- Rapid Convergence: The Binary Cross-Entropy (BCE) loss exhibits a sharp decline within the first 20 epochs, indicating that the GCN efficiently captures the most prominent topological features of the patent semantic graph in the early stages of training.
- Asymptotic Stability: The loss trajectory reaches an asymptotic floor around epoch 60, after which the gradients stabilize. The absence of significant stochastic fluctuations or late-stage divergence confirms that the all-MiniLM-L6-v2 SBERT embeddings provide a high-quality, linearly separable feature space that is highly compatible with the GCN's message-passing architecture.
- Structural Consistency: A notable observation in Fig. 11 is the high density of the individual batch curves (shown in light grey) relative

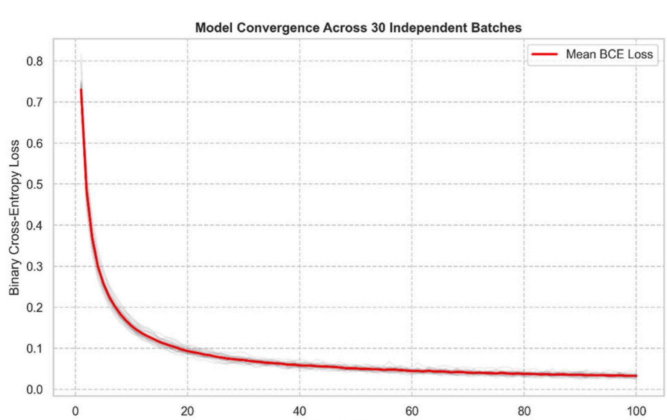

Fig. 11. Model convergence trajectory across 30 independent batches.

to the mean loss (red line). This indicates that despite the global random sampling of 5000 patents from the vast 233,122 CPC Class H population, the underlying latent semantic structure remains remarkably consistent.

- Generalization Performance: The smooth convergence profile, coupled with a near-zero loss floor and a Mean Accuracy of 0.955, demonstrates that the model effectively generalizes across the entire technological domain without suffering from severe overfitting or localized biases.

In summary, the convergence analysis validates that the proposed framework is not only accurate but also computationally robust, requiring minimal training iterations to reach a reliable decision state for Strategic Topological Value (STV) assessment.

5. Discussion and conclusion

This study set out to address a persistent challenge in R&D management: the accurate, early-stage valuation of patents amidst the "data overload" of the AI era. By proposing a multimodal deep learning framework that integrates Sentence-BERT (SBERT) with Graph Convolutional Networks (GCN), we have demonstrated that patent value is not merely a function of textual content but is intrinsically tied to its topological position within the technological network.

5.1. Discussion of empirical findings

5.1.1. The dominance of structural information

The most striking finding of this study is the significant performance gap between the proposed GCN framework (Accuracy: 0.955; F1-score: 0.956) and the "topology-blind" baselines (MLP Accuracy: 0.825; RF Accuracy: 0.794). This discrepancy provides strong empirical support for the "Structural Capital Hypothesis" in innovation economics.

While SBERT embeddings effectively capture the semantic nuances of an invention, they fail to account for the systemic importance of the technology. Text-only models treat each patent as an independent entity, making them blind to the latent connectivity that defines influence. In contrast, the GCN's neighborhood aggregation mechanism allows the model to learn that a patent's value is derivative of its relational context—effectively identifying that an invention connected to other central "hubs" is itself a critical node. This confirms that where a patent stands in the technological manifold is a more powerful predictor of strategic value than what its disclosure explicitly says.

5.1.2. Optimal topology and the over-smoothing trade-off

A key methodological contribution of this research is the Dynamic Thresholding mechanism. Our sensitivity analysis (Fig. 10) demonstrated that the model maintains high predictive stability (Accuracy $>$95%) across a wide range of target average degrees ($K \in [20, 30]$).

- **The "Goldilocks" Zone ($K = 25$):** The model achieves peak performance by capturing the most relevant semantic ties without introducing excessive noise. This density provides sufficient structural context for the GCN to identify Strategic Topological Value (STV) with near-deterministic precision.
- **Semantic Dilution and Noise:** As the graph density increases beyond the optimal threshold, we observe the onset of "Over-smoothing" or "Semantic Dilution." By forcing nodes to connect to weaker, less relevant neighbors, the GCN aggregates increasingly noisy features, which dilutes the discriminatory power of the core structural signals. This finding offers a practical guideline for future graph-based patent analytics: sparser, high-fidelity networks are computationally more efficient and predictively more robust for large-scale industrial deployment.

5.2. Managerial implications

For Chief Technology Officers (CTOs), IP Managers, and Tech Investors, this study offers an actionable roadmap for data-driven innovation strategy:

5.2.1. Early-stage valuation for real-time decision making

Traditional valuation relies on forward citations [6], forcing managers to wait 3-5 years. Our framework eliminates this bottleneck by providing a "Zero-Lag" proxy for value. Managers can deploy this model on newly published patent applications to immediately identify "hidden gems"—high-potential technologies that are structurally positioned to become future hubs. This capability is invaluable for securing licensing deals and identifying M&A targets ahead of the broader market.

5.2.2. Data-driven portfolio pruning

Large corporations often struggle with "zombie patents" that consume maintenance fees without providing strategic value. Our model provides a granular probability score for every asset. R&D managers can utilize these scores to execute a "Pruning Strategy", identifying low-centrality patents for abandonment or divestment, thereby reallocating capital toward high-STV projects with greater long-term impact.

5.3. Limitations and future research

Despite its robust performance, this study has limitations that outline future research avenues:

- **Abstracts vs. Full Claims:** To maximize the Signal-to-Noise Ratio (SNR), we utilized patent abstracts as the primary input. While abstracts provide a condensed technical summary, future iterations could integrate claim-based linguistic features to capture the specific legal boundaries of the invention.
- **Temporal Dynamics:** Our current model constructs a static snapshot of the technological landscape. Future research should explore Dynamic Graph Neural Networks (DGNNs) to model the evolution of a patent's value trajectory as the technology frontier shifts over time.

5.4. Conclusion

This research successfully addresses the persistent "time-lag" and "cold-start" dilemmas in early-stage patent valuation. By introducing an integrated framework that synthesizes Sentence-BERT (SBERT) embeddings with Graph Convolutional Networks (GCN), we have facilitated a paradigm shift in valuation—moving from a conventional "content-only" analysis to a holistic Structural Positioning assessment, operationalized through the novel concept of Strategic Topological Value (STV).

Our findings provide definitive answers to the research questions established in this study.

- Measuring Patent Value: We demonstrate that an invention's significance is fundamentally embedded within its topological coordinates in a latent semantic technological network. By employing a neighborhood aggregation mechanism (specifically optimized at $K = 25$), our GCN model effectively quantifies the positional advantage an invention holds relative to its technological peers.
- Overcoming Algorithmic Bias: Our approach successfully mitigates the "topology-blind" bias inherent in traditional NLP and machine learning models. Empirical validation across 30 independent experimental runs—utilizing global random samples from the extensive CPC Section H (Electricity) population ($N = 233,122$)—reveals that the inclusion of structural topology elevates predictive accuracy from 0.825 (text-only MLP) to a robust 0.955 (GCN), supported by a near-deterministic AUC of 0.993.

The academic and practical contributions of this research are significant. Scholarly, we substantiate the hypothesis that the structural context of an invention is a more decisive predictor of its ultimate impact than its isolated textual disclosure. We further clarify that STV captures the technological nexus of a patent, a metric distinct from, yet complementary to, its legal defensibility. Practically, this framework provides R&D managers and CTOs with a "Zero-Lag" decision support tool. By identifying high-potential technologies immediately upon publication, organizations can optimize their patent portfolios and competitive intelligence without the standard multi-year wait for citation data to mature.

Finally, we acknowledge the limitations that define future research trajectories. To maximize the Signal-to-Noise Ratio (SNR) for semantic encoding, this study utilized patent abstracts as the primary textual input. While we recognize that patent claims constitute the legally binding scope of protection, our results suggest that the technical density of abstracts is sufficient—and perhaps more efficient—for identifying a patent's relative position within a semantic network. Future studies could explore hybrid architectures that integrate claim-based features to refine assessments of legal quality. Additionally, while our focus on CPC Section H provides robust evidence in high-tech domains, exploring cross-domain generalizability and the integration of dynamic temporal graphs will be essential to track the evolution of Structural Positioning as the global technological landscape shifts.

Declaration of competing interest

I have no conflicts of interest to disclose.

Data availability

The data that support the findings of this study are available in a public repository.

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